

Applying ODE-Based Pharmacokinetic Models to Optimize Antihistamine Dosing Schedules

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1 Background

The field of study with respect to dosing schedules is currently dictated by pharmacokinetics and pharmacodynamics. The former concern the actions of drugs on the body, including the four stages: absorption, distribution, metabolism, and excretion of drugs [11]; the latter concern the actions of drugs on the patient’s body [11]. Currently, the dosing schedule of a drug reflects the “general public”, which excludes those at the extremes of age, weight, as well as people who are pregnant, and those with genetic disorders [9]. Major factors that contribute to the variation of the absorption schedules include these aforementioned attributes [12] and vary by absorption method [10].

This motivates the research question: *“How can ODE-based pharmacokinetic models be used to optimize antihistamine dosing schedules that maintain drug concentrations within a therapeutic window?”*

We believe drug dosing schedules are a suitable topic for modelling due to the following:

- Little modelling has been developed within the field, and in most cases, systems of ODEs with only a few parameters are applied, including compartmental models [1], and PBPK models [4].
- Both health [6] and ethical [8] concerns imply clinical trials are inappropriate for this field.
- Factors affecting dosing schedules such as age, gender, and weight are suitable factors to parameterize and model ODEs.

2 Mathematical and Computational Context

Pharmacokinetics can be described by multi-compartmental ordinary differential equations [7] by simplifying predictions of instantaneous drug concentration in a space in the body, increasing in accuracy with more models [5]. Three-compartment models are widely used as they account for the rapid initial distribution phase, and the subsequent slower elimination phase, which is sustained by the gradual release of the drug from tissue storage sites [13]. As the drug is administered in the central compartment (blood plasma), it distributes homogeneously while diffusing into a peripheral compartment (tissues) [3]. When elimination is incorporated, the system exhibits a bi-exponential decline. The third compartment (fat) serves as a site of drug accumulation and subsequently functions as a secondary source, gradually releasing the drug back into circulation.

This ODE system below reveals the dynamics of compartments, and address whether the concentration remains within the therapeutic window. By analyzing the amount and frequency of dosing from these plots, the optimal dosing schedule can be identified. Dosing quantities that produce the longest half-life can be inferred using their respective gradients [13]. Additionally, simulating repeated dosing demonstrates how concentration converges to a steady-state pattern, reflecting the balance between drug input and elimination.

Maintaining drug concentration within the therapeutic window can be framed as an optimization problem and evaluated using computational approaches. One technique is gradient-based optimization. This algorithm moves along the gradient (rate of concentration change) of an objective function within the therapeutic window range until the minimum is found, analyzing how small changes in dosing quantity and frequency affect drug concentration [2]. This algorithm can be applied to drug concentration-time graphs for single and multiple doses, as well as phase plots of different compartments to reveal interactions and highlight time delays.

ODE modeling, stability analysis, optimization, and computational simulation form a structured framework for investigating how dosing schedules can be optimized to keep antihistamine concentrations within the therapeutic window.

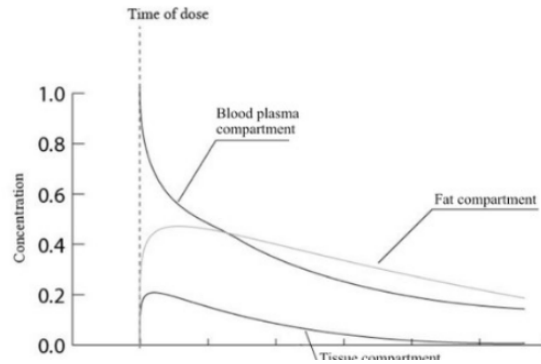


Figure 1: Three-compartment concentration phase plot [13]

3 Objectives Each Week

Week 1: Background research on pharmacokinetic concepts and compartmental ODE models.

Week 2: Implement one and two-compartment models, verifying against analytical half-life formulae from single dose concentration-time plots.

Week 3-4: Extend to three-compartment models with slower dynamics and simulate repeated dosing schedules to analyze how concentration approaches steady state.

Week 5-7: Implement computational optimization framework and test different dosing amounts and frequencies to identify the optimal schedule.

Week 8-9: Apply real data to simulation for analysis.

4 Expected Outcomes

The expected outcome of this research project is to not only understand how antihistamines are absorbed from the bloodstream by tissues, but also to optimize a model for dosing schedules, as not much research has been conducted in this area of pharmacokinetics. In general, we expect to see a general global decay in the concentrations of the antihistamines in the bloodstream as time passes when calculating and optimizing these models. Furthermore, we aim to eventually use these models and optimize dosing schedules using ODEs for various antihistamines and will parameterize these optimized models using a variety of factors, such as age, weight, and gender.

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